★ = our contribution

= standard tool / baseline

Minimum control cost under stabilization

finite-state systems (BCN as the motivating special case)

A. Undiscounted ($\lambda = 1$)

baseline

Single-sink shortest path

one reverse Dijkstra sweep

J^* + proper in-tree μ^*

exact, $O(|E| + |R|\log|R|)$

(standard; cycles harmless)

★ B. Discounted ($\lambda \in (0,1)$)

proper-feedback $J_\lambda^* \geq V_\lambda^*$

std VI / Floyd return V_λ^* (wrong)

★ Beneficial-cycle criterion

Lemma: $g_{\text{loop}} < (1-\lambda^\ell)\, J_\lambda^{\text{tail}}$

$\Rightarrow$ exactly when $V_\lambda^* < J_\lambda^*$

$\Rightarrow$ why reverse Dijkstra fails

Shared pipeline (three graph transforms)

Step 1. Restriction

keep $R = \{x : x \to^* x_e\}$, drop the rest;

states outside R have $J_\lambda^* = \infty$. $O(|E|)$

Step 2. Condensation

collapse each strongly-connected team to a point

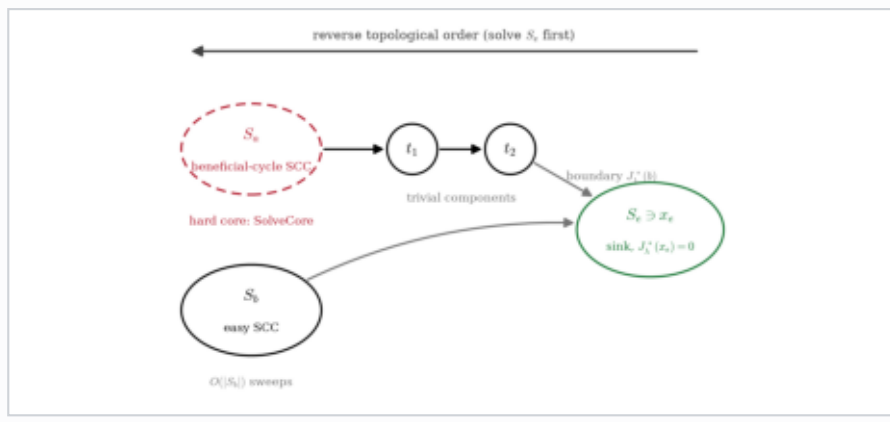
$\Rightarrow$ a DAG, giving a reverse topological order

★ Step 3. Folding $\to \omega$

fold solved downstream into one absorbing ω,

exit $e(x) = g + \lambda J_\lambda^*(b)$ (single λ);

each component $\to$ discounted simple path



paper Fig.: condensation DAG (Step 2) + reverse-topo order

TODO (detection step)

which case? test whether the V_λ^*-greedy policy is improper

improper $\Rightarrow$ beneficial cycle $\Rightarrow$ SolveCore; else $V_\lambda^* = J_\lambda^*$

Across components: solve in reverse topological order over the condensed DAG (boundary $J_\lambda^*(b)$ propagates)

within each component, exactly one of:

Trivial node (DAG)

no internal arc

$J_\lambda^* = \min$ exit

(one Bellman step)

SCC, no beneficial cycle

label-correcting (not Dijkstra)

walk = simple path $\Rightarrow$ in-tree

exact cost = edges $\times$ iterations,

iterations = feedback depth (small)

★ SCC with beneficial cycle

core is NP-hard (Ham. path)

SolveCore: subset DP

$O(2^{|S|}\,|S|\,M)$, exp. only in $|S|$

+ depth-parametrized scheme

Symbols

x_e	target equilibrium (cost-free sink)	μ^*	optimal proper feedback (in-tree)	ω	virtual absorbing sink node
λ	discount factor, $\lambda \in (0,1)$	R	stabilizable set (can reach x_e)	$e(x)$	exit weight $g + \lambda J_\lambda^*(b)$
g	stage cost $g(x,u) \geq 0$	M	number of control inputs	b	downstream boundary state ($\notin S$)
J^*	min undiscounted stab. cost	$\lvert E\rvert$	number of arcs (transitions)	C, ℓ	non-target cycle and its length
J_λ^*	min discounted stab. cost (proper)	D	optimal-feedback depth ($\leq \lvert R\rvert$)	g_{loop}	one-pass discounted cost of C
V_λ^*	unconstrained optimum (VI / Bellman)	S, $\lvert S\rvert$	a component (SCC) and its size	J_λ^{tail}	proper discounted cost-to-go